

of the tax year suppose there is a loss. Rebalancing would say buy more equities. The tax timing option says sell AA index Fund. The investor proceeds to sell AA Index Fund, thus realizing the loss which is deductible, and then buys the same amount and more of BB Market ETF. Economically, AA Index Fund is equivalent to BB Market ETF. For tax purposes, they are different. Doing this, the investor has both rebalanced and harvested the tax losses.

4.2. TAX-EFFICIENT ASSET ALLOCATION

The asset allocation and asset location problems are potentially complex given the potential interactions and dynamic tax strategies. There are two common approaches:

1. To specify the joint asset allocation and asset location problem and to solve the two problems simultaneously. An example is Dammon, Spatt, and Zhang (2004).
2. To ignore the interaction between the two decisions and implement ad hoc rules. Usually the asset allocation problem is solved first and then, taking the asset allocation as a given, the asset location problem is solved.

The second method is by far the more popular in the financial services industry. For example, Vanguard says:²¹

Consider placing the most tax-efficient holdings (such as tax-managed, index, and tax-exempt funds) in your taxable accounts while holding tax-inefficient investments in your tax-advantaged accounts—such as IRAs or employer plans.

It turns out that industry is right (partly). The optimal location problem is separable from the asset allocation problem, and the overall allocation problem can be done by considering only taxable accounts, rather than a considerably more complex joint problem over taxable and tax-deferred accounts. Jennifer Huang (2008), in a paper that was her PhD dissertation, was the first to show this.

Huang's model aligns with popular industry advice, with an extremely important caveat which I detail below. Investors place assets with the highest tax rates in the tax-deferred account to maximize the tax benefit. Bonds are highly taxed because coupons are taxed as income (at 36.9%). So these are placed in the tax-deferred account. Stocks with no dividends are subject only to the long-term capital gains tax (at 23.8%). So stocks are placed in taxable accounts.

²¹ <https://personal.vanguard.com/us/insights/taxcenter/how-to-be-tax-savvy-investor>